Risk Data Scientist, Take-Home Case Study

Candidate Brief: Investigating a Reported Account Takeover at FinFlow

Suggested time: 2–4 hours | AI tools: allowed and encouraged | Presentation: 45-minute panel session at your next interview

1. Welcome

Thank you for taking on this exercise. It is a realistic, open-ended investigation, the kind of problem our team works on daily. FinFlow is a fictional company and the data is fully synthetic, but the scenario reflects real patterns in payments risk.

There is no single right answer. We care about how you approach the problem, what you find, how you use your tools, including AI, and how you communicate your conclusions. Be honest about limitations and show your reasoning. Plan for 2–4 hours of work; please do not exceed 4.

2. The Scenario

> *FinFlow is a FinTech company that offers accounts-payable services for small and mid-sized businesses: customer organizations use the FinFlow platform to pay their vendors and suppliers via ACH, wire, virtual card, international wire, and check. Vendors are added to the platform by the customer, payments go through configurable approval chains, and FinFlow's risk-operations team reviews transactions flagged as suspicious.*
>
> *One day, one of FinFlow's customer businesses reaches out in alarm: they believe their FinFlow account has been taken over, they are seeing vendor payments they never initiated.*

You are a risk data scientist at FinFlow. You have been given all transaction and login activity for the last 90 days for the reporting customer and the surrounding customer population, including risk-operations review results and fraud labels.

Your task: investigate what happened, how far it reaches, and what FinFlow should do about it. Then present your findings.

3. How FinFlow Works (business context)

- **Organizations and users:** each customer organization has one or more users who log in to initiate and approve payments. Users are identified by user_id; the dataset does not include job titles or roles.
- **Vendors:** payees are added to an organization's vendor list by its users, along with the vendor's bank details. Bank details can later be changed.
- **Payments:** a payment specifies a vendor, an amount, and a transfer type. Organizations configure approval chains (e.g., payments above a threshold need a second approver). Some users hold permissions that allow bypassing the approval chain; when that happens it is recorded on the transaction.
- **Risk operations:** FinFlow's ops team reviews a subset of transactions that are flagged by rules, models, or customer reports. They do not review everything.

4. Label Documentation

The dataset contains three label fields. Understanding how each is produced matters:

- **dispute_flag**: set when the customer organization disputes a transaction. Disputes are customer-initiated: they only happen when the customer notices a problem, and they can lag the transaction by days or weeks.
- **ops_review_result**: the outcome of a risk-operations review. Values: `cleared`, `low_risk`, `high_risk`, `confirmed_fraud`. This field is only populated for transactions ops actually reviewed; unreviewed transactions are null.
- **txn_fraud_label**: the final adjudicated fraud outcome for the transaction, determined after full investigation. This is the most complete ground-truth signal in the dataset.

Think carefully about what each label does and does not tell you, and when each becomes available in real life.

5. Data Dictionary

Two CSV files, joined on `session_id`.

5.1 transactions.csv

Field	Type	Description
`txn_id`	id	Unique transaction identifier
`org_id`	categorical	The FinFlow customer organization that initiated the payment
`user_id`	id	The user within the organization who initiated the transaction
`txn_ts`	timestamp	Transaction time (UTC)
`amount`	numerical	Payment amount
`currency`	categorical	Payment currency
`transfer_type`	categorical	Payment rail: ACH, wire, virtual card, international wire, or check
`vendor_id`	categorical	The vendor (payee) receiving the payment
`vendor_add_date`	timestamp	When this vendor was added to the organization's vendor list
`vendor_bank_country`	categorical	Country of the vendor's receiving bank account
`is_new_bank_account`	true/false	Whether the vendor's bank details were added or changed within the last 7 days
`approval_bypassed`	true/false	Whether the transaction skipped the organization's normal approval chain (possible for users holding elevated permissions)
`session_id`	id	The login session the transaction was initiated from; joins to logins.csv
`dispute_flag`	true/false	Whether the customer organization disputed this transaction (see label documentation below)

Field	Type	Description
ops_review_result	categorical	Outcome of risk-operations review, if the transaction was reviewed (see label documentation below)
txn_fraud_label	true/false	Final adjudicated fraud outcome (see label documentation below)

5.2 logins.csv

Field	Type	Description
session_id	id	Unique session identifier; joins to transactions.csv
user_id	id	The user who logged in
org_id	categorical	The organization the user belongs to
login_ts	timestamp	Session start time (UTC)
device_id	categorical	Fingerprint of the device used for the session
ip_country	categorical	Country the session IP geolocates to
ip_region	categorical	State/region the session IP geolocates to
is_vpn_or_proxy	true/false	Whether the session IP was flagged as a VPN, proxy, or hosting provider
is_new_device	true/false	Whether this is the first time this device was seen for this user
mfa_triggered	true/false	Whether a multi-factor authentication challenge was issued
mfa_result	categorical	Outcome of the MFA challenge (passed / failed / n.a.)
login_success	true/false	Whether the login attempt succeeded

6. Deliverables

1. **Your notebook / code**: whatever you used for the analysis. It should run top to bottom.
2. **A presentation**: slides or a notebook walkthrough, your choice. You will present to a small panel in a dedicated 45-minute session: about 1–2 minutes introducing yourself (your background and what you enjoy working on), about 10 minutes on your findings, then questions from the panel for the remainder. Suggested coverage for the findings portion: how you processed the data; what you found (with visualizations); the model or scoring approach you built and how you evaluated it; what you recommend FinFlow do next.
3. **An AI-usage log**: required, lightweight (bullet points are fine), in four parts: (1) how you used AI to research and understand the problem; (2) how you used AI for data processing, coding, and model building; (3) how you used AI to prepare your presentation story and slides; (4) how you verified AI output was correct, include at least one example where the AI was wrong or you disagreed with it, and what you did.

7. Ground Rules & Tips

- **AI tools are allowed and encouraged**: Claude, ChatGPT, Gemini, Copilot, Cursor, anything you like. We want to see how you work with them. Transparency helps you: the AI-usage log and your ability to discuss it are part of the assessment.
- **The work must be yours to defend.** The take-home review is a dedicated 45-minute panel session: you introduce yourself for 1–2 minutes, present your findings for about 10 minutes, and the panel asks questions for the remainder about your choices, your code, and your findings, including “what if” variations.
- **Document your assumptions**: and anything in the data you found surprising or suspicious.
- **This brief is self-contained.** Everything you need is in this document and the dataset. If something seems ambiguous, make a reasonable assumption, document it, and proceed; working with imperfect information is part of the job. Contact your recruiting coordinator only for logistics issues, such as a file that will not download.

Good luck, and enjoy the investigation!